

Partial Derivatives

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Let us consider a function of two variables $z = f(x, y)$

Definition The **partial derivatives** of f are the functions defined by

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}, \quad \text{the partial derivative wrt } x$$

and

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}, \quad \text{the partial derivative wrt } y$$

Notation If $z = f(x, y)$ then

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

and

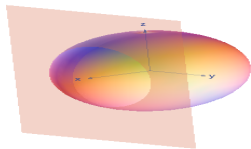
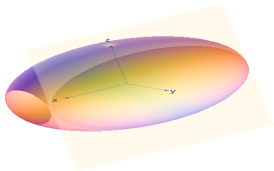
$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Now let us study the geometrical meaning of the partial derivatives.

Consider a surface given by $z = f(x, y)$ and $P(a, b, c)$ be a point on this surface. Then $f_x(a, b)$ is the slope of the tangent line to the intersection of the surface and the plane $y = b$ at $P(a, b, c)$.

Similarly, $f_y(a, b)$ is the slope of the tangent line to the intersection of the surface and the plane $x = a$ at $P(a, b, c)$.

Consider the surface $x^2 + 2y^2 + 3z^2 = 10$. Then $f_x(2, 1)$ and $f_y(2, 1)$ are:



How to calculate the partial derivatives?

For $z = f(x, y)$, when we compute

- $\frac{\partial f}{\partial x}$ we treat y as a constant and differentiate $f(x, y)$ wrt x
- $\frac{\partial f}{\partial y}$ we treat x as a constant and differentiate $f(x, y)$ wrt y

The partial derivatives can be interpreted as rates of change: $z = f(x, y) \implies \frac{\partial z}{\partial x} =$ the rate of change of z wrt x when y is fixed. Similarly, $\frac{\partial z}{\partial y} =$ the rate of change of z wrt y when x is fixed.

Example 1. Find the first partial derivatives of

- $f(x, y) = 2x^3 + 2y^2 - \frac{2x}{y} + \tan^{-1}(x^2 - y^3)$
- $z = e^{x^2 \cos(y)} + \ln(4x - 5y + 11)$
- $z = (-7y + 5x)^9 - \sqrt{\tan x + 3y + 9}$

Solution.

$$f(x, y) = 2x^3 + 2y^2 - \frac{2x}{y} + \tan^{-1}(x^2 - y^3)$$

$\Rightarrow$

$$f_x = 6x^2 - \frac{2}{y} + \frac{2x}{1 + (x^2 - y^3)^2}$$

$$f_y = 4y + \frac{2x}{y^2} - \frac{3y^2}{1 + (x^2 - y^3)^2}$$

Example 2 Find the first partial derivatives of

○ $f(x, y) = \ln(2x^3 - y) + 3y^2 - \frac{2x}{y}$

Example 3. Find $\frac{\partial f(x,y)}{\partial x}$ and $\frac{\partial f(x,y)}{\partial y}$ if $f(x,y) = \int_{2x}^{y^3} \tan(t^5) dt$

Solution

Solution

Recall the FTC:

$$\frac{d}{dx} \left(\int_{\alpha(x)}^{\beta(x)} f(t) dt \right) = f(\beta(x)) \cdot \beta'(x) - f(\alpha(x)) \cdot \alpha'(x)$$

$$f(x,y) = \int_{2x}^{y^3} \tan(t^5) dt \Rightarrow$$

$$f_y = \tan((y^3)^5) \cdot 3y^2$$

$$f_x = -\tan((2x)^5) \cdot 2$$

Example 4. Find the first partial derivatives of

○ $z = e^{x^2 \cos(y)} + \ln(4x - 5y + 11)$

Solution

$$f(x, y) = e^{x^2 \cos(y)} + \ln(4x - 5y + 11)$$

$\Rightarrow$

$$f_x = e^{x^2 \cos(y)} 2x \cos(y) + \frac{4}{4x - 5y + 11}$$

$$f_y = -e^{x^2 \cos(y)} x^2 \sin(y) - \frac{5}{4x - 5y + 11}$$

Example 5. Find the first partial derivatives of

○ $z = (-7y + 5x)^9 - \sqrt{\tan x + 3y + 9} + \tan^{-1}(x^2 - y^3)$

Obviously, one can take the partial derivative of a partial derivative. This leads us to the concept of higher derivatives or (mixed derivatives)

Definition The second partial derivatives of $z = f(x, y)$ are given by

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$ means that

we first differentiate wrt x then wrt y

$f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$ means that

we first differentiate wrt y then wrt x

Clairaut's Theorem If f_{xy} and f_{yx} are both continuous around a point (a, b) then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Remark. For the function

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

we have

$$f_{xy}(0, 0) = -1 \neq f_{yx}(0, 0) = 1.$$

Example Consider

$$f(x, y) = \begin{cases} \frac{x^3y - xy^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, we have

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0,$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

Using the quotient rule, we can easily show that for $(x, y) \neq (0, 0)$

$$f_x(x, y) = \frac{x^4y + 4x^2y^3 - y^5}{(x^2 + y^2)^2} \quad \text{and} \quad f_y(x, y) = \frac{x^5 - 4x^3y^2 - xy^4}{(x^2 + y^2)^2}$$

$$f_{xy}(0, 0) = \frac{\partial f_x}{\partial y} = \lim_{h \rightarrow 0} \frac{f_x(0, h) - f_x(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{-\frac{h^5}{h^4} - 0}{h} = -1$$

$$f_{yx}(0, 0) = \frac{\partial f_y}{\partial x} = \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^5}{h^4} - 0}{h} = 1$$

Exercise 6. Find all second partial derivatives of

◦ $f = 2x^5y^9 - 3 \tan(ax + by), \quad a, b \in \mathbb{R}$

◦ $u = \frac{s}{t^2 + 1}$

Solution

$$u_s = \frac{1}{t^2 + 1}, \quad u_t = \frac{-2st}{(t^2 + 1)^2}$$

$$u_{st} = \left(\frac{1}{t^2 + 1} \right)_t = \frac{-2t}{(t^2 + 1)^2} = u_{ts}$$

Exercise 7. Find

- f_{xyx} if $f = x^3y^2 - y^3 + x^4$
- $f_{xyyx}(0.4, 0.8)$ if $f = \sin^{-1}(x + y)$

Example 8. Use implicit differentiation to find all second partial derivatives of

$$2x^2 + y^2 + 5z^2 = 1 \quad (*)$$

Solution.

$$\text{Diff } (*) \text{ wrt } x \rightarrow 4x + 10z z_x = 0 \rightarrow z_x = -\frac{2x}{5z}$$

$$\text{Diff } (*) \text{ wrt } y \rightarrow 2y + 10z z_y = 0 \rightarrow z_y = -\frac{y}{5z}$$

$$\begin{aligned} z_{xy} &= \left(-\frac{2x}{5z} \right)_y = \left(-\frac{2x}{5} \right)_y \cdot \left(\frac{1}{z} \right)_y = \left(-\frac{2x}{5} \right) \cdot \left(\frac{-z_y}{z^2} \right) \\ &= \left(-\frac{2x}{5} \right) \cdot \left(\frac{\frac{y}{5z}}{z^2} \right) = \left(-\frac{2x}{5} \right) \cdot \left(\frac{y}{5z^3} \right) \\ &= -\frac{2xy}{25z^3} \end{aligned}$$

Example 4. Use implicit differentiation to find all second partial derivatives of

$$2x^2 + y^2 + 5z^2 + 3xyz = 1.$$

Final Remarks

Continuity vs Partial Derivatives!

We know that for a function $y = f(x)$, If $f(x)$ is differentiable at a point $x = a$, then it is also continuous at the same point. In other words, differentiability implies continuity. However, this **not true** for $z = f(x, y)$.

Example 5. The function

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

is not continuous at $(0, 0)$ even though both $f_x(0, 0)$ and $f_y(0, 0)$ exist.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0,$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

$\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist because along $y = mx$ the limit becomes

$$\lim_{x \rightarrow 0} \frac{mx^2}{x^2 + m^2x^2} = \frac{m}{1 + m^2}.$$

We can extend the idea of a partial derivative for functions with three or more variables.

Example 5. Let $w = f(x, y, z) = ye^{x^2y^5z^3}$. Find w_x , w_y and w_z .

Solution.

$$f(x, y, z) = ye^{x^2y^5z^3}$$

$\implies$

$$f_x = 2xy^6z^3e^{x^2y^5z^3}$$

$$f_y = e^{x^2y^5z^3} + 5x^2y^4z^3e^{x^2y^5z^3}$$

$$f_z = 3x^2y^6z^2e^{x^2y^5z^3}$$